Order of Operations with Radicals

Question Bank - Additional 4.06

Name_____
Date_____
Period_____

Show your work. Circle your final answer.

For questions 1 through 6, evaluate.

1. $30 - 4 \cdot 6 + 5$

2. $48 \div 8 \cdot 2$

3. $7 + 3(9 - 5)^2$

4. $\frac{5^2 + 11}{3^2 - 3}$

5. $-5^2 + (-5)^2$

6. $\sqrt{169 - 25} + 3^{-1}$

For questions 7 and 8, evaluate. Each expression has more than one grouping symbol.

7. Evaluate each expression.

a. $5(7.42-(2.6)^2)+\sqrt[3]{64}$

b. $\frac{3^3-3}{4^2}-\sqrt{\frac{5}{9}-\frac{4}{9}}$

c. $18\left(\left(\frac{1}{3}\right)^2+2^{-1}\right)$

8. Evaluate each expression.

a. $[5+(3-7)^2]\cdot 2^2$

b. $\frac{(-5)^3}{5^2}-\sqrt{121}$

c. $\sqrt[3]{-8}\cdot\left(\frac{1}{2}\right)^{-3}$

Working backwards.

9. Find the value of n that makes each statement true.

a. $n + 5 \cdot 3 = 27$

b. $(n-2)^2 = 64$, where n is greater than 2

c. $\frac{n+6}{3^2} = 4$

Reading someone else's work.

10. Liesel says that $10 - 4 + 3 = 3$, because addition comes before subtraction. Is she right? Explain, and give the correct value.

11. Which is the value of $25 - 3(2)^3$?

- ○ A. 7
- ○ B. −191
- ○ C. 1
- ○ D. 6,859

12. Select ALL of the expressions whose value is 9.

- ☐ A. $\left(\frac{1}{3}\right)^{-2}$
- ☐ B. $36 \div 2 \cdot 2$
- ☐ C. $4 + 6 - 1$
- ☐ D. $\sqrt{81}$
- ☐ E. -3^2
- ☐ F. $\frac{7^2 - 4}{5}$